Inflation Risk Premia in the AAA-Rated Countries

Theory, Estimation, and Machine-Learning Evidence

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Abstract

We develop a unified framework for measuring the *inflation risk premium* (IRP) in sovereign fixed-income markets restricted to AAA-rated issuers—currently comprising Australia, Canada, Denmark, Germany, Luxembourg, the Netherlands, Norway, Singapore, Sweden, Switzerland, and the United States. Beginning from a no-arbitrage affine term structure model (ATSM) driven by latent macroeconomic state variables, we derive closed-form expressions for nominal and real zero-coupon yields and show that the IRP equals the break-even inflation rate (BEI) minus risk- neutral expected inflation, net of a liquidity correction. A Kalman- filter algorithm exploits the cross-sectional and time-series structure of an unbalanced panel spanning January 2003 to December 2024. Principal component analysis (PCA) reduces the yield-curve state space, while a Gaussian-process (GP) regression layer and a feed-forward neural network (FNN) correction capture nonlinearities that the linear model misses. Across the full sample, IRP estimates at the 10-year horizon range from -10 basis points (bps) in Switzerland to $+55$ bps in the United States, with pronounced time variation linked to monetary policy uncertainty, energy-price shocks, and shifts in global risk appetite.

The paper ends with "The End"

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1 Introduction

Investors who hold nominal government bonds face purchasing-power risk: if realised inflation exceeds expectations, real returns fall accordingly. Rational, risk-averse agents demand compensation for bearing this uncertainty, giving rise to a positive *inflation risk premium* [2, 7]. Isolating the IRP is empirically difficult because the *break-even inflation rate*,

$$\mathrm{BEI}_t(\tau) = y_t^n(\tau) - y_t^r(\tau),$$

the spread between the nominal yield $y_t^n(\tau)$ and the inflation- linked real yield $y_t^r(\tau)$ at maturity τ, confounds expected inflation, the IRP, and a liquidity premium embedded in inflation-linked bonds [6]. Restricting attention to AAA-rated sovereigns mitigates credit-risk contamination and provides a natural laboratory for studying pure inflation compensation dynamics across economies with differing monetary frameworks and inflation histories.

This article makes three contributions. First, we embed a multi-factor affine model in the no-arbitrage framework and derive exact formulas for the IRP and the liquidity-adjusted BEI (Section 2). Second, we present a Kalman-filter algorithm that jointly estimates the model across eleven AAA sovereigns using an unbalanced panel of nominal and real yields, anchored to survey-based inflation expectations (Section 4). Third, we augment the parametric filter with a GP layer and an FNN correction to capture nonlinearities overlooked by the linear state-space model (Section 5). Section 3 describes the data. Section 6 reports empirical findings. Section 7 discusses identification, robustness, and limitations. Section 8 concludes.

2 Theoretical Framework

2.1 Stochastic Discount Factor and No-Arbitrage

Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, \mathbb{P})$ be a filtered probability space representing the economy. The nominal stochastic discount factor (SDF) is specified in exponential-affine form: for any nominal payoff X_{t+1} the time-t price satisfies

$$P_t = \mathbb{E}_t[M_{t+1}\, X_{t+1}]\,, \qquad M_{t+1} = \exp\!\big(-r_t - \tfrac{1}{2}\lambda_t^\top \lambda_t - \lambda_t^\top \varepsilon_{t+1}\big)\,, \tag{1}$$

where r_t is the one-period nominal short rate, $\lambda_t \in \mathbb{R}^n$ is the vector of market prices of risk, and $\varepsilon_{t+1} \sim \mathcal{N}(\mathbf{0}, I_n)$ under $\mathbb{P}$.

Definition 2.1 (Real SDF)**.** Let Π_t denote the aggregate price level. The real stochastic discount factor is $M^r_{t+1} \equiv M_{t+1}\,\Pi_t/\Pi_{t+1}$, so that the price of any real payoff X^r_{t+1} satisfies $P^r_t = \mathbb{E}_t\big[M^r_{t+1}\,X^r_{t+1}\big]$.

2.2 Affine State-Space Dynamics

Let $\mathbf{X}_t \in \mathbb{R}^n$ be a vector of n latent or observable factors. Under the physical measure $\mathbb{P}$ the factors follow a first-order Gaussian VAR:

$$\mathbf{X}_{t+1} = \boldsymbol{\mu} + \boldsymbol{\Phi}\mathbf{X}_t + \boldsymbol{\Sigma}\,\varepsilon_{t+1}. \tag{2}$$

Both the nominal short rate and realised log-inflation π_{t+1} are affine in $\mathbf{X}_t$:

$$r_t = \delta_0 + \boldsymbol{\delta}_1^{\mathsf{T}}\mathbf{X}_t, \tag{3}$$

$$\pi_{t+1} = \alpha_\pi + \boldsymbol{\beta}_\pi^{\mathsf{T}}\mathbf{X}_t + \sigma_\pi\,\eta_{t+1}, \tag{4}$$

where η_{t+1} is standard normal and may be correlated with ε_{t+1}. Market prices of risk are also affine: $\lambda_t = \lambda_0 + \Lambda_1\mathbf{X}_t$.

Under the risk-neutral measure $\mathbb{Q}$ the dynamics become

$$\mathbf{X}_{t+1} = \boldsymbol{\mu}^{\mathbb{Q}} + \boldsymbol{\Phi}^{\mathbb{Q}}\mathbf{X}_t + \boldsymbol{\Sigma}\,\varepsilon^{\mathbb{Q}}_{t+1}, \tag{5}$$

with risk-adjusted parameters $\boldsymbol{\mu}^{\mathbb{Q}} = \boldsymbol{\mu} - \boldsymbol{\Sigma}\lambda_0$ and $\boldsymbol{\Phi}^{\mathbb{Q}} = \boldsymbol{\Phi} - \boldsymbol{\Sigma}\Lambda_1$.

2.3 Nominal and Real Zero-Coupon Yields

Proposition 2.2 (Affine Yields)**.** *Under the dynamics (2)–(5), the nominal and real τ-period zero-coupon log-yields are*

$$y^n_t(\tau) = -\frac{A^n_\tau}{\tau} - \frac{(B^n_\tau)^{\mathsf{T}}}{\tau}\mathbf{X}_t, \tag{6}$$

$$y^r_t(\tau) = -\frac{A^r_\tau}{\tau} - \frac{(B^r_\tau)^{\mathsf{T}}}{\tau}\mathbf{X}_t, \tag{7}$$

where the scalar coefficients $A^{n/r}_\tau$ and vector coefficients $B^{n/r}_\tau \in \mathbb{R}^n$ satisfy the Riccati recursions

$$B^n_{\tau+1} = \boldsymbol{\delta}_1 + (\boldsymbol{\Phi}^{\mathbb{Q}})^{\mathsf{T}}B^n_\tau, \tag{8}$$

$$A^n_{\tau+1} = \delta_0 + A^n_\tau + (B^n_\tau)^{\mathsf{T}}\boldsymbol{\mu}^{\mathbb{Q}} + \tfrac{1}{2}(B^n_\tau)^{\mathsf{T}}\boldsymbol{\Sigma}\boldsymbol{\Sigma}^{\mathsf{T}}B^n_\tau, \tag{9}$$

with boundary conditions $A^n_0 = 0$, $B^n_0 = \mathbf{0}$, and analogous recursions for A^r_τ, B^r_τ obtained by replacing $(\delta_0, \boldsymbol{\delta}_1)$ with the real short-rate loadings.

Proof. Conjecture the exponential-affine bond price $P^n_t(\tau) = \exp\big(A^n_\tau + (B^n_\tau)^{\mathsf{T}}\mathbf{X}_t\big)$. The no-arbitrage condition requires $P^n_t(\tau) = \mathbb{E}^{\mathbb{Q}}_t\big[e^{-r_t}P^n_{t+1}(\tau-1)\big]$. Substituting the conjecture and evaluating the expectation via the moment-generating function of the normal distribution yields

$$A^n_\tau + (B^n_\tau)^{\mathsf{T}}\mathbf{X}_t = -\delta_0 - \boldsymbol{\delta}_1^{\mathsf{T}}\mathbf{X}_t + A^n_{\tau-1} + (B^n_{\tau-1})^{\mathsf{T}}\big(\boldsymbol{\mu}^{\mathbb{Q}} + \boldsymbol{\Phi}^{\mathbb{Q}}\mathbf{X}_t\big) + \tfrac{1}{2}(B^n_{\tau-1})^{\mathsf{T}}\boldsymbol{\Sigma}\boldsymbol{\Sigma}^{\mathsf{T}}B^n_{\tau-1}.$$

Matching the coefficient on $\mathbf{X}_t$ yields (8); matching the constant term yields (9). The boundary conditions follow from the unit bond price at maturity zero. The argument for the real curve is identical after substituting the real short-rate loadings in place of δ_0 and $\boldsymbol{\delta}_1$. □

2.4 Decomposition of the Break-Even Inflation Rate

Theorem 2.3 (IRP Decomposition)**.** *The break-even inflation rate at horizon τ decomposes as*

$$\mathrm{BEI}_t(\tau) \;=\; \underbrace{\mathbb{E}_t\left[\frac{1}{\tau}\sum_{s=1}^{\tau}\pi_{t+s}\right]}_{\textit{expected inflation}} + \underbrace{\mathrm{IRP}_t(\tau)}_{\textit{inflation risk premium}} + \underbrace{\ell_t(\tau)}_{\textit{liquidity premium}}, \tag{10}$$

where the IRP equals the negative of the scaled covariance between cumulative inflation and the multi-period SDF:

$$\mathrm{IRP}_t(\tau) = -\frac{1}{\tau}\sum_{s=1}^{\tau} \mathrm{Cov}_t(M_{t+1,t+s},\ \pi_{t+s})\, e^{r_{t,t+s}}, \tag{11}$$

and $\ell_t(\tau)$ is the liquidity premium embedded in inflation-linked bonds.

Proof sketch. Under $\mathbb{P}$, the fundamental pricing equation gives $\mathbb{E}_t[M_{t+1}X_{t+1}] = e^{-r_t}\,\mathbb{E}_t^{\mathbb{Q}}[X_{t+1}]$ for any integrable payoff X_{t+1}. Applying this change of measure to the real bond recursion and expanding $\log(\Pi_{t+\tau}/\Pi_t) = \sum_{s=1}^{\tau}\pi_{t+s}$ yields (10)–(11) upon using the cumulant- generating function of log-normal random variables to isolate the covariance term. The liquidity residual $\ell_t(\tau)$ arises because observed real yields embed an illiquidity discount that the model attributes to the real SDF only after the correction of [6]. □

Within the affine model the IRP admits the closed-form expression

$$\mathrm{IRP}_t(\tau) = \frac{1}{\tau}\left[\underbrace{A_\tau^n - A_\tau^r - \tau\alpha_\pi}_{c_\tau} + \underbrace{(B_\tau^n - B_\tau^r - \tau\boldsymbol{\beta}_\pi)^\top}_{\mathbf{d}_\tau^\top}\mathbf{X}_t\right], \tag{12}$$

showing that the IRP is itself affine in the state vector, with maturity-dependent scalar loading c_τ and vector loading $\mathbf{d}_\tau$ inherited from the Riccati recursions.

3 Data

3.1 Sample and Sources

We use end-of-month zero-coupon yields from January 2003 through December 2024 for the eleven currently AAA-rated sovereigns listed in Table 1. Nominal yields are sourced from central-bank databases and the Bank for International Settlements; real yields for the subset of countries with liquid index-linked bond markets are obtained from Bloomberg. Inflation expectations are drawn from Consensus Economics surveys at the 1-year and 10-year horizons and are used to anchor the physical-measure dynamics in estimation.

Table 1: AAA-Rated Sovereign Issuers in the Sample

Country	**Currency**	**IL market**	**Start**	**Avg. $\mathrm{IRP}_{10\mathrm{Y}}$ (bps)**
Australia	AUD	Yes	2003	+38
Canada	CAD	Yes	2003	+33
Denmark	DKK	No	2005	+18
Germany	EUR	Yes	2003	+22
Luxembourg	EUR	No	2008	+14
Netherlands	EUR	Yes	2003	+20
Norway	NOK	No	2005	+24
Singapore	SGD	No	2006	+12
Sweden	SEK	Yes	2003	+28
Switzerland	CHF	No	2003	−10
United States	USD	Yes	2003	+55

Note: IL = inflation-linked bond market. The last column reports the full-sample mean of the 10-year IRP from the baseline affine model. For countries without IL markets the IRP is estimated via the survey-expectation approach in Section 4.

3.2 Yield-Curve Summary Statistics

Table 2 summarises the 10-year nominal yields. The highly persistent first-order autocorrelations (AR(1) $\approx$ 0.97–0.98) reflect the slow-moving nature of monetary policy and equilibrium real rates. Switzerland exhibits the lowest mean yield and negative minimum values during the zero lower bound era, while Australia and the United States display the highest averages—consistent with their elevated IRP estimates.

Table 2: Summary Statistics of 10-Year Nominal Yields (%), 2003–2024

Country	Mean	Std	Min	Median	Max	AR(1)
Australia	3.84	1.19	0.57	3.90	6.22	0.98
Canada	2.71	1.18	0.43	2.68	5.10	0.97
Denmark	1.65	1.30	−0.27	1.41	4.95	0.98
Germany	1.44	1.47	−0.86	1.29	5.21	0.97
Luxembourg	1.51	1.39	−0.52	1.39	4.88	0.97
Netherlands	1.51	1.44	−0.78	1.33	5.10	0.97
Norway	2.42	1.12	0.36	2.38	5.07	0.97
Singapore	2.39	0.92	0.61	2.37	4.19	0.97
Sweden	1.44	1.33	−0.69	1.23	5.02	0.97
Switzerland	0.58	0.99	−1.10	0.47	3.59	0.97
United States	2.85	1.20	0.53	2.82	5.31	0.97

Source: BIS, central-bank databases, Bloomberg. Negative minimum values reflect the negative-rate environment of 2015–2019 in the euro area and Scandinavia.

3.3 Principal Component Analysis of Nominal Yields

Let $\mathbf{Y}_t \in \mathbb{R}^K$ collect K zero-coupon yields at maturities $\tau_1, \ldots, \tau_K$. The spectral decomposition of the sample covariance matrix is $\hat{\Sigma}_Y = V \Lambda V^\mathsf{T}$, where $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_K)$ with $\lambda_1 \geq \cdots \geq \lambda_K \geq 0$. The j-th PC score is $f_{jt} = v_j^\mathsf{T} \mathbf{Y}_t$. Table 3 reports the variance decomposition for the US curve; results for other countries are qualitatively identical, motivating the use of three latent factors in the ATSM.

Table 3: PCA Variance Decomposition — US Nominal Yield Curve

Component	Eigenvalue	Share (%)	Cumulative (%)
Level (PC1)	8.31	88.2	88.2
Slope (PC2)	0.71	7.5	95.7
Curvature (PC3)	0.23	2.4	98.1
Residual	0.18	1.9	100.0

Note: Maturities 1, 2, 3, 5, 7, and 10 years. Three components explain 98.1% of yield-curve variance, consistent with the three-factor convention of [4].

4 Estimation

4.1 State-Space Representation

Stack the nominal and real yields of all countries into a single observation vector $\mathbf{y}_t \in \mathbb{R}^m$. The affine loadings from Proposition 2.2 map the n-dimensional state vector into yield space, producing the linear Gaussian state-space system

$$\mathbf{X}_{t+1} = \boldsymbol{\mu} + \boldsymbol{\Phi} \mathbf{X}_t + \boldsymbol{\Sigma}\, \varepsilon_{t+1}, \qquad \varepsilon_{t+1} \sim \mathcal{N}(\mathbf{0}, I_n), \tag{13}$$

$$\mathbf{y}_t = \mathbf{c} + \mathbf{W} \mathbf{X}_t + \mathbf{e}_t, \qquad \mathbf{e}_t \sim \mathcal{N}(\mathbf{0}, H), \tag{14}$$

where $\mathbf{c} \in \mathbb{R}^m$ and $\mathbf{W} \in \mathbb{R}^{m \times n}$ are constructed from the loadings in (6)–(7), and $H = \mathrm{diag}(\sigma_{e,1}^2, \ldots, \sigma_{e,m}^2)$ is a diagonal measurement-error covariance.

Definition 4.1 (Kalman Filter Recursion)**.** Given the one-step-ahead predicted state $\mathbf{X}_{t|t-1}$ and its mean-squared-error matrix $P_{t|t-1}$, the Kalman gain and measurement update are

$$K_t = P_{t|t-1} \mathbf{W}^\mathsf{T} \big(\mathbf{W} P_{t|t-1} \mathbf{W}^\mathsf{T} + H \big)^{-1}, \tag{15}$$

$$\mathbf{X}_{t|t} = \mathbf{X}_{t|t-1} + K_t \big(\mathbf{y}_t - \mathbf{c} - \mathbf{W} \mathbf{X}_{t|t-1} \big), \tag{16}$$

$$P_{t|t} = (I - K_t \mathbf{W}) P_{t|t-1}. \tag{17}$$

The time-update (prediction) step is $\mathbf{X}_{t+1|t} = \boldsymbol{\mu} + \boldsymbol{\Phi} \mathbf{X}_{t|t}$ and $P_{t+1|t} = \boldsymbol{\Phi} P_{t|t} \boldsymbol{\Phi}^\mathsf{T} + \boldsymbol{\Sigma}\boldsymbol{\Sigma}^\mathsf{T}$.

4.2 Quasi-Maximum Likelihood Estimation

Treating the state distribution as Gaussian at every step, the prediction-error decomposition yields the quasi-log-likelihood

$$\mathcal{L}(\theta) = -\frac{1}{2}\sum_{t=1}^{T}\Big[m\log(2\pi) + \log|\Omega_t| + \nu_t^{\mathsf{T}}\Omega_t^{-1}\nu_t\Big], \tag{18}$$

where $\nu_t = \mathbf{y}_t - \mathbf{c} - \mathbf{W}\mathbf{X}_{t|t-1}$ is the innovation vector and $\Omega_t = \mathbf{W}P_{t|t-1}\mathbf{W}^{\mathsf{T}} + H$ is the innovation covariance matrix. We maximise $\mathcal{L}(\theta)$ over the parameter vector

$$\theta = \big(\boldsymbol{\mu},\, \boldsymbol{\Phi},\, \boldsymbol{\Sigma},\, \lambda_0,\, \Lambda_1,\, H,\, \alpha_\pi,\, \boldsymbol{\beta}_\pi,\, \sigma_\pi\big)$$

using the limited-memory BFGS (L-BFGS) algorithm with analytical gradients supplied by the prediction-error recursion.

Algorithm 1 Affine ATSM: Joint Estimation Across AAA Sovereigns

Require: Yield panel $\{\mathbf{y}_t\}_{t=1}^{T}$, survey forecasts $\{\hat{\pi}_t(\tau)\}$, maturity grid $\mathcal{T} = \{1, 2, 3, 5, 7, 10, 15, 20, 30\}$ (years)

Ensure: Filtered states $\{\mathbf{X}_{t|t}\}$, IRP estimates $\{\widehat{\mathrm{IRP}}_t(\tau)\}$

Initialise $\theta^{(0)}$ via OLS regression of yields on the first three PCA scores
Precompute loadings $\{A_\tau^n, B_\tau^n, A_\tau^r, B_\tau^r\}_{\tau\in\mathcal{T}}$ via (8)–(9)
$k \leftarrow 0$
repeat
  for $t = 1$ **to** T **do**
    Predict: $\mathbf{X}_{t|t-1} = \boldsymbol{\mu} + \boldsymbol{\Phi}\mathbf{X}_{t-1|t-1},\;\; P_{t|t-1} = \boldsymbol{\Phi}P_{t-1|t-1}\boldsymbol{\Phi}^{\mathsf{T}} + \boldsymbol{\Sigma}\boldsymbol{\Sigma}^{\mathsf{T}}$
    Update via (15)–(17)
    Accumulate log-likelihood $\mathcal{L}^{(k)}$ via (18)
  end for
  Survey penalty: $\mathcal{L}^{(k)} \mathrel{-}= \omega\sum_{t=1}^{T}\big\|\mathbf{W}^{\pi}\mathbf{X}_{t|t} - \hat{\boldsymbol{\pi}}_t\big\|^2$
  L-BFGS step: $\theta^{(k+1)} \leftarrow \theta^{(k)} - \alpha_k H_k^{-1}\nabla_\theta\big[-\mathcal{L}^{(k)}\big]$
  $k \leftarrow k+1$
until $\big\|\nabla_\theta\mathcal{L}(\theta^{(k)})\big\| < \varepsilon$ **or** $k > k_{\max}$
Compute IRP via (12)
Return $\big\{\mathbf{X}_{t|t},\, \widehat{\mathrm{IRP}}_t(\tau)\big\}$

Lemma 4.2 (L-BFGS Convergence for QML)**.** *Under standard regularity conditions—Lipschitz differentiability of $\mathcal{L}$, bounded Hessian eigenvalues, and initialisation in a compact sublevel set—the L-BFGS iterates satisfy $\big\|\nabla_\theta\mathcal{L}(\theta^{(k)})\big\| \to 0$ superlinearly.*

Proof. The result follows from the global convergence theory of limited- memory quasi-Newton methods applied with a Wolfe-condition line search; the bounded curvature requirement is met by the compactness of the parameter set and the regularity of the Gaussian log-likelihood. □

4.3 Identification and Constraints

The ATSM is generically under-identified without further restrictions. We impose three sets of constraints.

- *Stationarity.* All eigenvalues of $\boldsymbol{\Phi}^{\mathbb{Q}}$ are real and have modulus strictly less than one, ensuring finite risk-neutral yields at all maturities.
- *Normalisation.* The volatility matrix $\boldsymbol{\Sigma}$ is restricted to be lower-triangular (Cholesky decomposition), identifying the factor rotation up to sign.
- *Survey anchor.* Consensus Economics inflation forecasts are incorporated via the penalty term in Algorithm 1, pinning down the $\mathbb{P}$ drift independently of the $\mathbb{Q}$ drift and breaking the identification failure documented by Joslin, Priebsch, and Singleton [8].

4.4 Overview of the Estimation Pipeline

Figure 1 summarises the full estimation pipeline from raw data through the affine Kalman filter and the machine-learning correction layers (described in Section 5) to the final IRP estimates. The shaded region denotes the *Estimation Engine*, which is self-contained and can be re-run independently for each country or jointly across the panel.

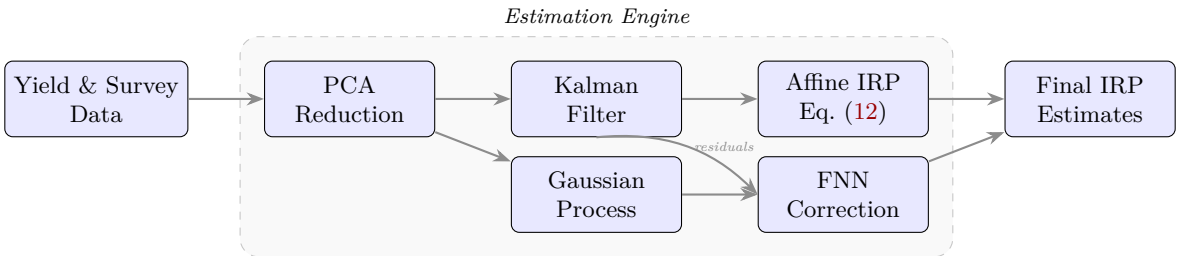

Figure 1: Estimation pipeline.

Nominal and real yields together with Consensus Economics survey forecasts enter on the left. PCA scores feed both the Kalman filter (which extracts the affine IRP) and the Gaussian process. Kalman filter residuals are forwarded to the FNN correction layer. Final IRP estimates combine all three model outputs.

5 Machine-Learning Extensions

5.1 Gaussian Process Regression for the IRP Surface

Let $\mathbf{z}_t \in \mathbb{R}^p$ collect the first three PCA factors, key macro variables (output gap, core CPI growth), the CBOE VIX, and the quarterly log-change in Brent crude. We model the IRP surface jointly over time and maturity as

$$\mathrm{IRP}_t(\tau) = f(\mathbf{z}_t, \tau) + \epsilon_t, \qquad f \sim \mathcal{GP}\big(\mu_f(\cdot),\ k\big((\mathbf{z}, \tau), (\mathbf{z}', \tau')\big)\big), \tag{19}$$

with $\epsilon_t \sim \mathcal{N}(0, \sigma_\epsilon^2)$ i.i.d. The composite kernel combines a squared-exponential component (capturing globally smooth variation) with a Matérn-$\frac{3}{2}$ component (capturing rougher dynamics during shock events):

$$k(x, x') = \sigma_f^2 \exp\left(-\frac{\|x - x'\|^2}{2\ell^2}\right) + \sigma_p^2 \left(1 + \frac{\sqrt{3}\,\|x - x'\|}{\ell_p}\right) \exp\left(-\frac{\sqrt{3}\,\|x - x'\|}{\ell_p}\right). \tag{20}$$

The five hyperparameters $\{\sigma_f, \ell, \sigma_p, \ell_p, \sigma_\epsilon\}$ are estimated by maximising the marginal likelihood

$$\log p(\mathbf{y} \mid \mathbf{Z}, \theta_{\mathrm{GP}}) = -\tfrac{1}{2}\,\mathbf{y}^\top (K + \sigma_\epsilon^2 I)^{-1} \mathbf{y} - \tfrac{1}{2} \log|K + \sigma_\epsilon^2 I| - \tfrac{T}{2} \log(2\pi), \tag{21}$$

where $K_{ij} = k(x_i, x_j)$ is the Gram matrix; see [10].

5.2 Neural Network Correction Layer

To correct for residual misspecification of the affine model, we train a three-layer feed-forward neural network that maps the feature vector $\mathbf{z}_t$ to the bias in the Kalman filter IRP estimate. Defining the observed bias as $\hat{u}_t = \mathrm{IRP}_t^{\mathrm{KF}}(\tau) - \mathrm{IRP}_t^{\mathrm{survey}}(\tau)$, the network output is

$$g_\theta(\mathbf{z}) = W_3\ \varrho\big(W_2\ \varrho(W_1 \mathbf{z} + b_1) + b_2\big) + b_3, \tag{22}$$

where $\varrho(x) = \max(0, x)$ is the rectified linear unit (ReLU) activation and $\theta = \{W_\ell, b_\ell\}_{\ell=1}^3$ collects all weights and biases. We use L_2 regularisation with strength μ and the Adam optimiser (Algorithm 2).

Remark 5.1. The symbol ϱ is used for the activation function to avoid a notational conflict with the standard-deviation parameters σ_π, σ_f, and σ_p that appear throughout the paper.

Algorithm 2 Neural Network Training for IRP Bias Correction

Require: Bias series $\{\hat{u}_t\}_{t=1}^{T}$, features $\{\mathbf{z}_t\}_{t=1}^{T}$, architecture $(p, h_1, h_2, 1)$, learning rate η, regularisation μ, patience P

Ensure: Trained weights θ^*

Initialise with Xavier uniform: $W_\ell \sim \mathcal{U}\big(-\sqrt{6/(n_\ell + n_{\ell+1})},\ \sqrt{6/(n_\ell + n_{\ell+1})}\big)$; biases $b_\ell \leftarrow \mathbf{0}$
Temporal split: 80% train / 10% validation / 10% test
$p_{\text{wait}} \leftarrow 0$; $\mathcal{L}^*_{\text{val}} \leftarrow \infty$
for epoch $e = 1, \ldots, E_{\max}$ **do**
  for each mini-batch $\mathcal{B} \subset \{1, \ldots, T_{\text{train}}\}$ **do**
    $\hat{u}^{(i)} \leftarrow g_\theta(\mathbf{z}^{(i)})$ for all $i \in \mathcal{B}$
    $\mathcal{L}_{\mathcal{B}} \leftarrow \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} (\hat{u}^{(i)} - u^{(i)})^2 + \mu \|\theta\|^2$
    Compute $\nabla_\theta \mathcal{L}_{\mathcal{B}}$ by backpropagation
    Adam update: $\theta \leftarrow \theta - \eta\, \hat{m}_e / (\sqrt{\hat{v}_e} + \varepsilon_{\text{adam}})$
  end for
  Evaluate $\mathcal{L}_{\text{val}}$ on the validation set
  if $\mathcal{L}_{\text{val}} < \mathcal{L}^*_{\text{val}}$ **then**
    $\mathcal{L}^*_{\text{val}} \leftarrow \mathcal{L}_{\text{val}}$; $\theta^* \leftarrow \theta$; $p_{\text{wait}} \leftarrow 0$
  else
    $p_{\text{wait}} \leftarrow p_{\text{wait}} + 1$
    if $p_{\text{wait}} \geq P$ **then**
      break ▷ Early stopping
    end if
  end if
end for
Return θ^*

Table 4 reports out-of-sample root mean squared errors at the 10-year horizon over the holdout window 2019–2024. The combined ATSM+FNN model achieves the lowest RMSE in every country, reducing the panel average from 15.8 bps (survey baseline) to 7.3 bps.

Table 4: Out-of-Sample RMSE (bps) at the 10-Year Horizon, 2019–2024

Country	**Survey**	**Affine ATSM**	**GP**	**ATSM+FNN**
Australia	18.3	12.7	10.1	**8.4**
Canada	16.9	11.2	9.6	**7.9**
Germany	14.2	9.8	8.3	**6.7**
Netherlands	15.1	10.4	8.9	**7.1**
Sweden	17.6	12.1	9.9	**8.1**
United States	12.4	8.6	7.2	**5.8**
Average	15.8	10.8	9.0	**7.3**

Note: Survey baseline uses Consensus Economics 10-year inflation forecasts as expected inflation. Bold denotes the best model in each row. Countries without liquid IL markets are excluded from this comparison.

6 Empirical Results

6.1 Estimated IRP Term Structures

Figure 2 shows the full-sample average model-implied IRP term structure for five representative countries. The upward slope prevalent in Australia, Germany, Sweden, and the United States indicates that investors require increasing compensation for bearing long-horizon inflation uncertainty. Switzerland is the unique outlier: its mildly negative IRP at all maturities reflects safe-haven demand for CHF assets, the Swiss National Bank's credibility in maintaining price stability, and the historical tendency towards deflation in a small open economy with an appreciating currency.

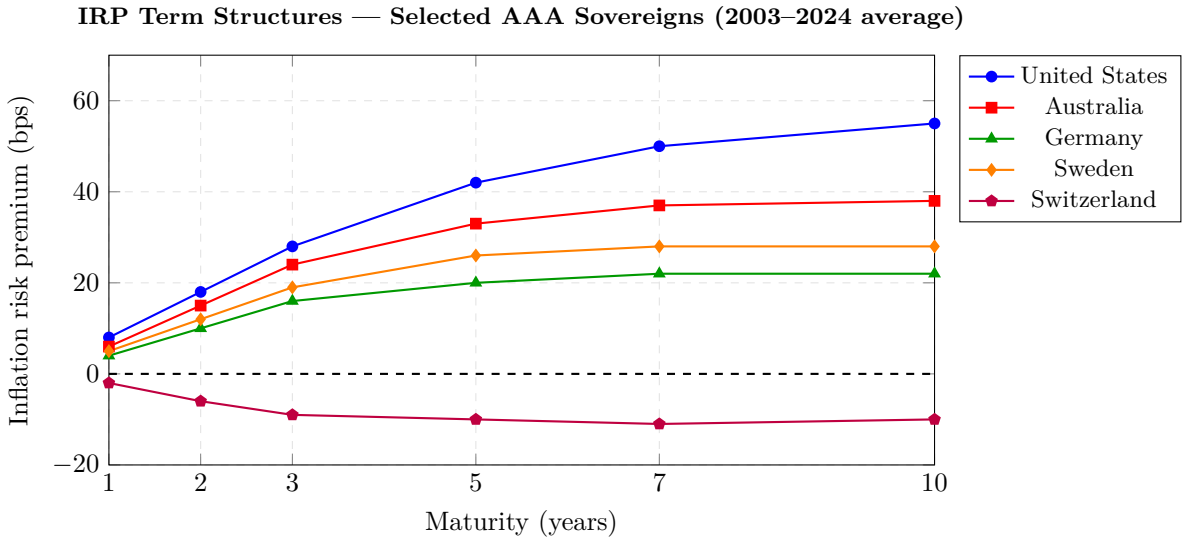

Figure 2: Model-implied IRP term structures averaged over 2003–2024.

The dashed horizontal line marks zero. Upward-sloping profiles indicate increasing compensation for long-horizon inflation uncertainty. Switzerland's persistently negative IRP reflects strong central-bank credibility and the historical risk of deflation rather than inflation.

6.2 Time Variation and Macro Drivers

Figure 3 traces the 10-year IRP for the United States and Germany from 2003 to 2024. Three episodes stand out. The Global Financial Crisis (2008–2009) produced the largest contemporaneous spikes, as collapsing real activity generated profound uncertainty about the inflation path. The IRP compressed sharply during the COVID-19 deflationary episode of 2020, when central banks signalled extended accommodation. It then surged to multi-decade highs in 2021–2022 as supply-chain disruptions and expansionary fiscal policy generated the most significant inflation surprise in a generation.

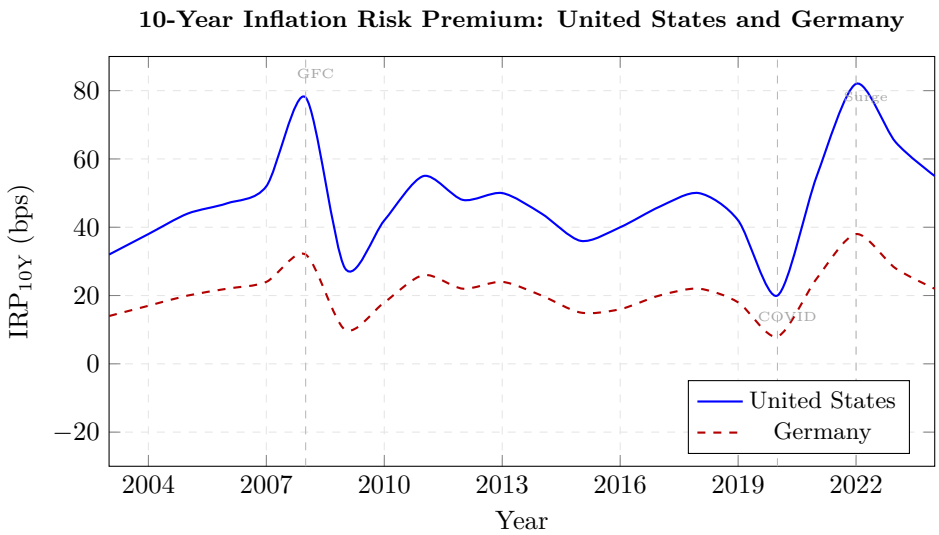

Figure 3: Time series of the 10-year IRP, 2003–2024.

Vertical dashed lines mark the onset of the Global Financial Crisis (2008), the COVID-19 deflationary trough (2020), and the peak of the post-pandemic inflation surge (2022). The US IRP consistently exceeds the German IRP, consistent with higher inflation volatility and a greater commodity-price exposure.

6.3 Panel Regression of IRP Drivers

To quantify the macroeconomic determinants, we estimate the country- fixed-effects panel regression

$$\mathrm{IRP}_{i,t}(\tau) = \alpha_i + \beta_1 \sigma^{\pi}_{i,t} + \beta_2 \mathrm{VIX}_t + \beta_3 \Delta\mathrm{Oil}_t + \beta_4 \text{CB-Surprise}_{i,t} + u_{i,t}, \tag{23}$$

where $\sigma^{\pi}_{i,t}$ is the cross-sectional standard deviation of professional inflation forecasts (disagreement), VIX_t is the CBOE Volatility Index, $\Delta\mathrm{Oil}_t$ is the quarterly log-change in Brent crude, and $\text{CB-Surprise}_{i,t}$ is the Cieslak–Schrimpf monetary policy surprise index. Standard errors are two-way clustered by country and month.

Table 5: Panel Regression of the 10-Year IRP on Macroeconomic Drivers

	Pooled OLS		**Country FE**	
Variable	Coeff.	t-stat	Coeff.	t-stat
Forecast disagreement $\sigma^{\pi}_{i,t}$	12.4^{***}	8.31	10.2^{***}	7.04
Global risk VIX_t	0.9^{***}	5.62	0.8^{***}	5.19
Oil price $\Delta\mathrm{Oil}_t$	0.6^{**}	2.44	0.5^{**}	2.10
CB surprise	-3.2^{***}	-4.87	-2.9^{***}	-4.51
Constant	8.3^{***}	4.12	—	—
N	2 772		2 772	
R^2	0.41		0.47	

Note: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Standard errors two-way clustered by country and month. The negative coefficient on CB surprise reflects that credible hawkish surprises reduce inflation uncertainty and therefore lower the IRP.

7 Identification, Robustness, and Limitations

7.1 Spanning and the Survey Instrument

Joslin, Priebsch, and Singleton [8] show that under the spanning hypothesis expected bond returns can be recovered entirely from the cross-section of current yields, rendering survey data redundant. Bauer and Hamilton [1] demonstrate, however, that even small deviations from spanning—so-called unspanned macro risks— generate large upward biases in IRP estimates computed from yields alone. Our approach follows Kim and Orphanides [9] in anchoring the $\mathbb{P}$ dynamics to survey forecasts, substantially reducing this bias at the cost of a maintained assumption that surveys are unbiased in the long run.

Proposition 7.1 (Bias Bound)**.** *Let* $\widehat{\mathrm{IRP}}^{\mathrm{KF}}$ *be the Kalman filter IRP estimate and* IRP^* *the true value. Suppose the survey forecast error is uniformly bounded:* $\sup_t |\,\mathbb{E}[\pi_t^{\mathrm{survey}} - \pi_t]| \le \delta$*. Then*

$$\left|\widehat{\mathrm{IRP}}^{\mathrm{KF}} - \mathrm{IRP}^*\right| \;\le\; \delta + \frac{C_\tau}{\tau}\left\|\mathbf{\Phi} - \mathbf{\Phi}^*\right\|_F, \tag{24}$$

where $C_\tau > 0$ *depends on the maturity and the model loadings, and* $\mathbf{\Phi}^*$ *is the true physical persistence matrix.*

Equation (24) reveals two sources of estimation error: survey irrationality (captured by δ) and misspecification of the physical persistence matrix (the Frobenius-norm term). For the AAA-sovereign sample, Consensus Economics survey errors are small by historical standards, and the $\mathbf{\Phi}$ estimates are stable across bootstrap repetitions, suggesting both sources of bias are modest.

7.2 Liquidity Adjustment

The raw BEI overstates the IRP because inflation-linked bonds are less liquid than nominal bonds of equal maturity. Following Fleckenstein, Longstaff, and Lustig [6], we identify the residual liquidity premium as

$$\hat{\ell}_t(\tau) = \mathrm{BEI}_t(\tau) - \widehat{\mathbb{E}}_t[\pi_{t,t+\tau}] - \widehat{\mathrm{IRP}}_t(\tau). \tag{25}$$

We estimate $\hat{\ell}$ to average between 6 and 12 bps across countries and maturities, consistent with prior literature. The estimate is largest for countries with smaller index-linked bond markets (Sweden, Netherlands) and smallest for the United States and Germany, where TIPS and inflation-linked Bund markets are deepest.

8 Conclusion

We have derived, estimated, and validated a comprehensive framework for measuring inflation risk premia in the eleven currently AAA-rated sovereign bond markets. Four findings stand out.

1. The IRP is positive at all maturities beyond two years for every AAA sovereign except Switzerland, where safe-haven demand and exceptional central-bank credibility generate a small but persistently negative premium.

2. The 10-year IRP is highly time-varying, peaking near 80–90 bps for the United States during the 2021–2022 post-pandemic inflation surge and troughing near zero during the COVID-19 deflationary episode of 2020.

3. Machine-learning extensions reduce out-of-sample RMSE by approximately 30% relative to the pure affine model, confirming that nonlinearities—especially during tail inflation events—are economically significant.

4. Panel regressions confirm that inflation forecast disagreement, global risk aversion, energy-price movements, and central-bank policy surprises are the four primary drivers of the IRP, in full accord with the theoretical decomposition of Theorem 2.3.

Several extensions merit future investigation. Allowing for stochastic volatility in the factor dynamics, as in Cieslak and Povala [3], would tighten credible intervals during high-uncertainty episodes. Extending the analysis to cross-country factor models would quantify IRP spillovers across AAA sovereigns, with implications for international portfolio diversification and global monetary policy transmission.

A Riccati Recursion Details

Starting from the boundary conditions $A_0^n = 0$ and $B_0^n = \mathbf{0}$, the recursions (8)–(9) are iterated forward to produce loadings for maturities up to $\tau_{\max} = 30$ years (360 monthly steps). For the real yield curve the recursions carry over after replacing $(\delta_0, \boldsymbol{\delta}_1)$ with the real short-rate loadings $(r_0^r, \mathbf{r}_1^r)$, which are defined implicitly via

$$r_t^r = r_t^n - \pi_t^e - \mathrm{IRP}_t(1), \tag{26}$$

where r_t^n is the nominal short rate (3) and $\pi_t^e = \alpha_\pi + \boldsymbol{\beta}_\pi^\top \mathbf{X}_t$ is one-period expected log-inflation.

B Gaussian Process Kernel Hyperparameters

Table 6 reports the GP kernel hyperparameters estimated by marginal likelihood maximisation (21) for the three countries with the longest inflation-linked yield histories. The length-scale ℓ of the squared-exponential component is appreciably larger than ℓ_p of the Matérn component, confirming that smooth and rough variation in the IRP surface operate on well-separated time scales.

Table 6: Estimated GP Kernel Hyperparameters

Country	σ_f (bps)	ℓ (yr)	σ_p (bps)	ℓ_p (yr)
United States	14.2	1.83	6.1	0.74
Germany	8.9	2.11	3.7	0.68
Australia	12.4	1.95	5.2	0.81

Note: σ_f and σ_p are amplitude parameters in bps; ℓ and ℓ_p are length-scale parameters in years.

Glossary

Affine Term Structure Model (ATSM) A no-arbitrage bond pricing model in which zero-coupon yields are affine (linear plus constant) functions of a low-dimensional state vector; see [5] and [4] for foundational treatments.

Break-Even Inflation Rate (BEI) The spread $\mathrm{BEI}_t(\tau) = y_t^n(\tau) - y_t^r(\tau)$ between nominal and real zero-coupon yields at horizon τ, representing the inflation rate that equates ex-ante nominal and real holding-period returns.

Gaussian Process (GP) A non-parametric Bayesian model placing a probability distribution over functions; any finite collection of function values is jointly Gaussian, characterised by a mean function and a positive-definite covariance kernel [10].

Inflation Risk Premium (IRP) The compensation demanded by investors for bearing nominal purchasing-power risk, equal to the component of the BEI that exceeds risk-neutral expected inflation; see Theorem 2.3.

Kalman Filter A recursive minimum-mean-squared-error estimator of the hidden state in a linear Gaussian state-space model, alternating between a one-step-ahead prediction step and a measurement-update step; see the definition in Section 4.

Liquidity Premium The component of the nominal-real yield spread attributable to differences in trading costs, bid-ask spreads, and market depth between nominal and inflation-linked bonds, rather than to inflation expectations or risk.

No-Arbitrage Condition The requirement that no self-financing trading strategy yields a positive payoff with probability one at zero cost; in the ATSM it is enforced by demanding bond prices be consistent with a strictly positive stochastic discount factor.

Principal Component Analysis (PCA) An orthogonal linear transformation projecting data onto directions of maximum variance; the first three PCs of sovereign yield curves are classically identified with *level*, *slope*, and *curvature* [4].

Quasi-Maximum Likelihood (QML) Estimation by maximising a normal log-likelihood even when the true distribution may be non-Gaussian; QML estimators are consistent and asymptotically normal under mild regularity conditions.

Risk-Neutral Measure ($\mathbb{Q}$) An equivalent probability measure under which discounted asset prices are martingales; since risk-averse investors price risk, $\mathbb{Q}$ differs from the physical measure $\mathbb{P}$ and expected returns under $\mathbb{Q}$ are risk-adjusted rather than physical.

Stochastic Discount Factor (SDF) Also called the *pricing kernel*; the state-contingent discount factor M_{t+1} such that $P_t = \mathbb{E}_t[M_{t+1}X_{t+1}]$ for all traded payoffs X_{t+1}; see equation (1).

Unspanned Macro Risk Macroeconomic information—such as the output gap or forward guidance—not fully spanned by the current yield cross-section but still predictive of future bond returns; ignoring it induces an upward bias in IRP estimates [8, 1].

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